EVEI LIGHT METER – Arduino Mega 2560 carrier shield

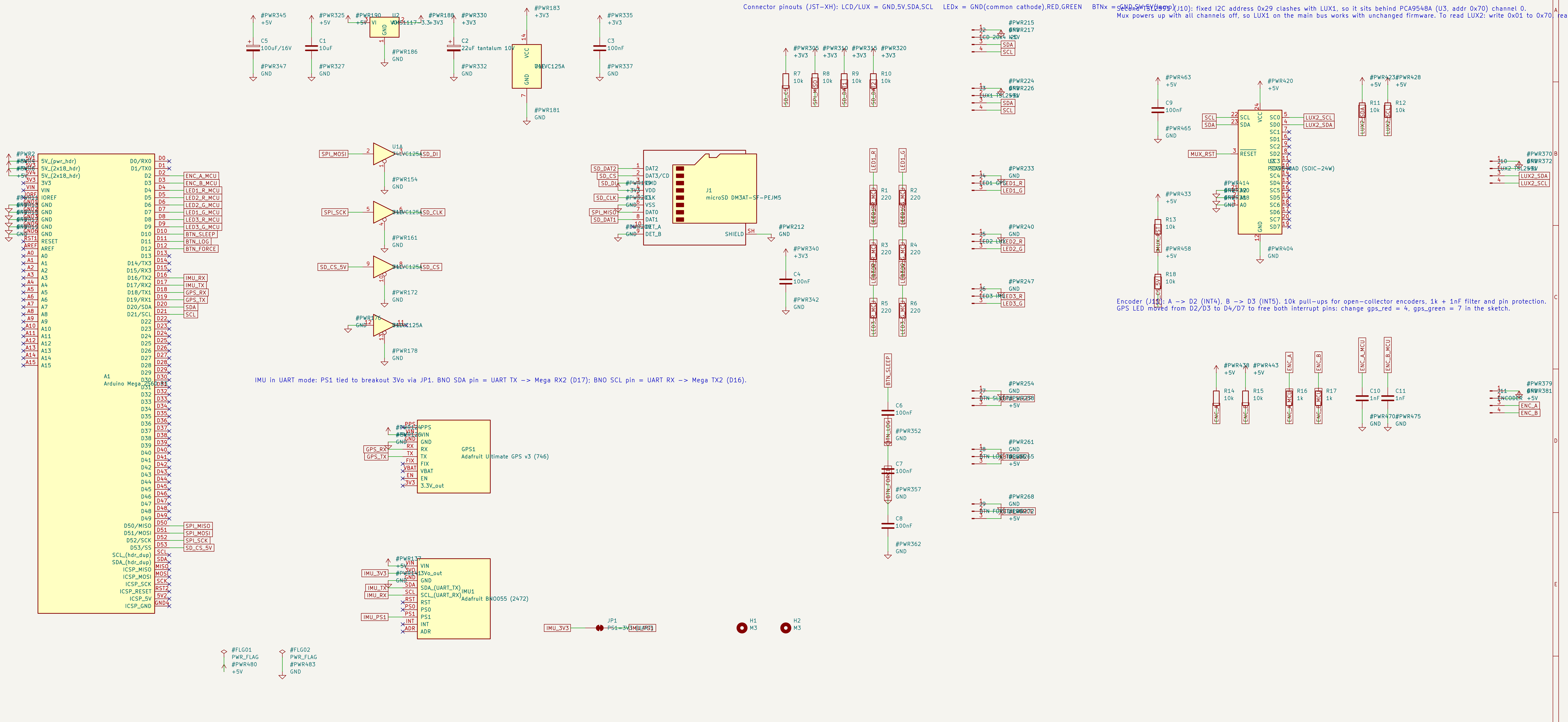
microSD: on-board socket, 3.3V LDO, 74LVC125 shifts MOSI/SCK/CS 5V–>3.3V; MISO (3.3V) goes straight to D50 (same as Adafruit 254 breakout).

Pin map from the team sketch (Lightmeter_Week8.ino): SD CS=D53 (HW SPI 50/51/52), GPS=Serial1 (D18/D19), IMU=Serial2 (D16/D17), I2C=D20/D21,
LEDs common–cathode to GND, active HIGH (setLED() in Week 5–8 sketches: YELLOW = both HIGH): GPS R=D4 G=D7, LUX R=D5 G=D6, IMU R=D8 G=D9; buttons to GND with INPUT_PULLUP: sleep D10, log toggle D11, force log D12.

Connector pinouts (JST–XH): LCD/LUX = GND,5V,SDA,SCL   LEDx = GND(common cathode),RED,GREEN   BTNx = GND,SW,5V(lamp)

REV B ADDITIONS

Second TSL2591 (J10): fixed I2C address 0x29 clashes with LUX1, so it sits behind PCA9548A (U3, addr 0x70) channel 0.
Mux powers up with all channels off, so LUX1 on the main bus works with unchanged firmware. To read LUX2: write 0x01 to 0x70, read, writ



A1
Arduino Mega 2560
U2
AMS1117–3.3
C5 100uF/16V
C1 10uF
C2 22uF tantalum 10V
C3 100nF
U1 74LVC125A
SPI_MOSI
SD_DI
SPI_SCK
SD_CLK
SD_CS_5V
SD_CS
J1
microSD DM3AT–SF–PEJM5
SHIELD
C4 100nF
R7 10k
R8 10k
R9 10k
R10 10k
LCD 20x4 I2C
LUX1 TSL2591
SDA
SCL
LED1_R
LED1_G
R1 220
R2 220
R3 220
R4 220
R5 220
R6 220
LED1 GPS
LED2 LUX
LED3 IMU
BTN_SLEEP
BTN_LOG
BTN_FORCE
C6 100nF
C7 100nF
C8 100nF
GPS1
Adafruit Ultimate GPS v3 (746)
GPS_RX
GPS_TX
IMU1
Adafruit BNO055 (2472)
IMU_3V3
IMU_TX
IMU_RX
IMU_PS1
JP1
PS1=3Vo
H1 M3
H2 M3
#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
C9 100nF
U3
PCA9548AD (SOIC–24W)
MUX_RST
LUX2_SCL
LUX2_SDA
R11 10k
R12 10k
R13 10k
R18 10k
LUX2 TSL2591
R14 10k
R15 10k
R16 1k
R17 1k
C10 1nF
C11 1nF
ENCODER
ENC_A
ENC_B
ENC_A_MCU
ENC_B_MCU
+5V
+3V3
GND

IMU in UART mode: PS1 tied to breakout 3Vo via JP1. BNO SDA pin = UART TX –> Mega RX2 (D17); BNO SCL pin = UART RX –> Mega TX2 (D16).

Encoder (J11): A –> D2 (INT4), B –> D3 (INT5). 10k pull–ups for open–collector encoders, 1k + 1nF filter and pin protection.
GPS LED moved from D2/D3 to D4/D7 to free both interrupt pins: change gps_red = 4, gps_green = 7 in the sketch.

Purdue EVEI
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